

PAUL MITCHELL

ARTIFICIAL INTELLIGENCE ENGINEER | AGENTIC SYSTEMS & AI SAFETY

Riverdale, MD | (413) 204-0493 | pjmitchell@wm.edu | pjmitchell7.github.io/shadowprompt

PROFESSIONAL SUMMARY

William & Mary M.S. in Computer Science candidate (3.95 GPA, AI/ML concentration) and clearable U.S. citizen targeting Booz Allen Hamilton's AI Engineering & Agentic Systems practice (Req R0244899). Bridges applied machine learning, multi-agent RAG architectures, and adversarial AI safety with hands-on systems engineering, enterprise FastAPI microservices, and 100+ node Linux infrastructure telemetry.

TECHNICAL SKILLS

Agentic AI & LLM Systems: LangGraph, LangChain, RAG architectures, vector databases (FAISS, Milvus), prompt engineering, LLM guardrails

Machine Learning & NLP: PyTorch, Hugging Face, DistilBERT, Sentence-BERT, BERTopic, LightGBM, scikit-learn, model evaluation

Backend & Cloud Engineering: Python, FastAPI, REST APIs, Docker, AWS (S3, EC2, Bedrock), Linux (Ubuntu), Git/GitLab CI, SQL

Systems & Observability: Prometheus, Grafana, telemetry instrumentation, NIST AI Risk Management Framework, automated evaluation

TECHNICAL & PROFESSIONAL EXPERIENCE

Data Analytics & AI Extern | Extern, Amazon-Sponsored Project | Remote Jun 2025 - Sep 2025

- Engineered end-to-end NLP pipelines combining fine-tuned DistilBERT ($F1 = 0.89$) and BERTopic clustering on ~20,000 unstructured text records, automating feature extraction and topic modeling.
- Built a predictive classification engine using LightGBM and deployed an interactive Streamlit application delivering real-time decision-support analytics and metric visualization for leadership.
- Prototyped an LLM retrieval insight interface querying structured text corpora with grounded citations.

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA Feb 2025 - Present

- Administer 100+ Linux instructional compute servers, configuring networking, DNS, and user isolation via Puppet.
- Deployed Prometheus and Grafana telemetry dashboards monitoring compute cluster utilization, hardware memory pressure, and node health, accelerating incident triage and infrastructure reliability.

KEY TECHNICAL PROJECTS

ShadowPrompt: Adversarial LLM Red-Teaming & Honey-Prompt Defense | pjmitchell7.github.io/shadowprompt 2026

- Engineered an adversarial fuzzer simulating steganographic token smuggling (zero-width payloads and prompt injection) against defense-grade LangGraph multi-agent RAG pipelines.
- Built an air-gapped pre-inference security proxy catching 99.2% of attacks in <1ms; engaged honey-prompt sandboxes to neutralize data exfiltration and enforced citation grounding under NIST AI Risk standards.

PromptShield: Tokenization-Aware Adversarial LLM Firewall | Python, FastAPI, scikit-learn 2026

- Engineered a pre-inference security proxy intercepting prompt injection and jailbreak attacks prior to model execution; benchmarked at 0.27-1.11 ms scoring latency with zero model retraining required.

EDUCATION

Master of Science in Computer Science, AI/ML Concentration | Expected Dec 2026 | GPA: 3.95

Bachelor of Science in Computer Science | William & Mary | Dec 2024

U.S. Citizen | Clearable for Secret / Top Secret / SCI Clearances